



JERUSALEM COLLEGE OF ENGINEERING
AN AUTONOMOUS INSTITUTION

Approved by AICTE, New Delhi, Affiliated to Anna University Chennai,
Accredited by NBA, New Delhi and Accredited by NAAC with "A" Grade.
Velachery Main Road, Narayanapuram, Pallikaranai, Chennai, Tamil Nadu - 600100.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

QUESTION BANK

Regulation	:	2023
Course Name	:	DATABASE MANAGEMENT SYSTEMS
Course Code	:	JCS2401
Year / Semester / Branch		II/IV/CSE
Unit No	:	1
Unit Title	:	DATA MODELS AND QUERYING
No. of Hours	:	9
Weightage in Exam (%)	:	20 %

PART A

Q.No.	Questions	Marks	BTL	CO
1.	Define database management system Database management system (DBMS) is a collection of interrelated data and a set of programs to access those data.	2	K1	CO1
2.	List any five applications of DBMS. • Banking, • Airlines • Universities, • Tele communication, • Finance, Sales, Manufacturing • Human resources	2	K1	CO1
3.	Outline the purpose of a DBMS. Data redundancy and inconsistency, Difficulty in accessing data, Data isolation, Integrity problems, Atomicity problems and Concurrent access anomalies.	2	K2	CO1
4.	Define instance and schema Instance: Collection of data stored in the data base at a particular moment is called an Instance of the database Schema: The overall design of the data base is called the data base schema.	2	K1	CO1
5.	Define the terms physical schema and logical schema. Physical schema:			



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	The physical schema describes the database design at the physical level, which is the lowest level of abstraction describing how the data are stored. Logical schema: The logical schema describes the database design at the logical level, which describes what data are stored in the database and what relationship exists among the data.	2	K1	CO1
6.	Classify the different types of data models used in database systems. • Hierarchical model • Network model • Relational model • Entity-Relationship (ER) model • Object-oriented model..	2	K2	CO1
7.	Outline the operations of relational algebra. Relational algebra is a procedural query language used to perform operations on relations (tables) in a database. It provides a set of fundamental operations to retrieve and manipulate data. These operations include selection, projection, union, set difference, and join. Relational algebra forms the theoretical foundation of SQL. It helps in understanding how queries are processed internally. It is important for designing efficient database queries and operations.	2	K2	CO1
8.	Categorize and define the different views of data in a database system. A database system has three levels of data abstraction: physical, logical, and view levels. The physical level describes how data is stored in storage devices. The logical level explains what data is stored and the relationships among data. The view level defines how users interact with the data. This separation helps in data independence and simplifies database management. Each user can have a customized view of the same database.	2	K2	CO1
9.	Why is a data dictionary important in DBMS? A data dictionary in DBMS is important because: • It stores metadata (data about data) such as table structure, data types, and constraints. • It ensures data consistency, integrity, and helps in efficient database management.	2	K2	CO1
10.	Compare SQL with relational algebra	2	K2	CO1



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	Feature	SQL (Structured Query Language)	Relational Algebra			
	Type of Language	Declarative (what to do)	Procedural (how to do)			
	User Friendliness	Easy and user-friendly	Less user-friendly			
	Purpose	Used for querying and managing databases	Theoretical foundation for DBMS			
	Usage	Used in real-world applications	Mainly used for understanding/query design			
	Execution	Directly executed in DBMS	Not directly executed			
	Level	High-level language	Low-level formal language			
	Operations	Uses SELECT, INSERT, UPDATE, DELETE	Uses Select, Project, Join, etc.			
11.	Differentiate between the database approach and the file-based approach based on their characteristics. File -based System. • Separation and isolation of data • Duplication of data • Incompatible file formats • Data dependence Database Approach: • Control of data redundancy • Data consistency • Sharing of data • Improved data integrity • Improved security			2	K2	CO1
12.	Name the disadvantages of file processing system? The file processing system has the following major disadvantages: □ Data redundancy and inconsistency □ Integrity Problems □ Security Problems			2	K1	CO1



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	□ Difficulty in accessing data □ Data isolation.											
13.	Interpret the features of a query language in DBMS. A query language is a language used to communicate with a database to retrieve and manipulate data. It allows users to perform operations such as inserting, updating, deleting, and retrieving data. Query languages provide a structured way to interact with databases without needing to know internal details. They help in efficient data access and management. The most commonly used query language is SQL (Structured Query Language), which is widely supported in relational databases.	2	K2	CO1								
14.	Mention the types of SQL command SQL commands are divided in to the following categories: 1. Data -definition language 2. Data manipulation language 3. Data Query language 4. Data control language 5. Data administration statements 6. Transaction control statements	2	K1	CO1								
15.	List the SQL domain Types SQL supports the following domain types: Char (n) , varchar (n), int , numeric (p,d) , float(n) , date.	2	K1	CO1								
16.	Give reason for using aggregate functions in SQL queries. Aggregate functions in SQL are used to perform calculations on multiple rows and return a single summarized value. They help in analyzing data efficiently, such as finding totals (SUM), averages (AVG), counts (COUNT), minimum (MIN), and maximum (MAX).	2	K2	CO1								
17.	<table><tr><td colspan="2">Differentiate static and dynamic SQL.</td></tr><tr><td>Static SQL</td><td>Dynamic SQL</td></tr><tr><td>In Static SQL, how database will be accessed is predetermined in the embedded SQL statement.</td><td>In Dynamic SQL, how database will be accessed is determined at run time.</td></tr><tr><td>SQL statements are compiled at compile time.</td><td>SQL statements are compiled at run time.</td></tr></table>	Differentiate static and dynamic SQL.		Static SQL	Dynamic SQL	In Static SQL, how database will be accessed is predetermined in the embedded SQL statement.	In Dynamic SQL, how database will be accessed is determined at run time.	SQL statements are compiled at compile time.	SQL statements are compiled at run time.	2	K2	CO1
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	<table><tr><td>Parsing, Validation, Optimization and Generation of application plan are done at compile time.</td><td>Parsing, Validation, Optimization and Generation of application plan are done at run time.</td></tr><tr><td>It is generally used for situations where data is distributed uniformly.</td><td>It is generally used for situations where data is distributed non uniformly.</td></tr></table>	Parsing, Validation, Optimization and Generation of application plan are done at compile time.	Parsing, Validation, Optimization and Generation of application plan are done at run time.	It is generally used for situations where data is distributed uniformly.	It is generally used for situations where data is distributed non uniformly.			
Parsing, Validation, Optimization and Generation of application plan are done at compile time.	Parsing, Validation, Optimization and Generation of application plan are done at run time.							
It is generally used for situations where data is distributed uniformly.	It is generally used for situations where data is distributed non uniformly.							
18.	Why does SQL allow duplicate tuples in a table or in a query result? If key constraint is not set on a relation every result in a relation will be considered as a tuple and hence SQL allows duplicate tuples in a table. Distinct keyword is used to avoid duplicate tuples in the result.	2	K2	CO1				
19.	List the DDL, DML, DCL and TCL commands. DDL COMMANDS: ● Create ● Alter ● Add ● Modify ● Drop ● Rename ● Drop DML Commands: ● Insert ● Select ● Update ● Delete DCL commands ● Grant -Provide access privilege to user ● Revoke -Get back access privilege from user TCL commands ● Commit ● Rollback Save point	2	K1	CO1				
20.	How does embedded SQL work?Give a sample syntax Embedded SQL works by incorporating SQL statements within a host programming language (such as C or Java). These SQL statements are processed by a precompiler and executed at runtime to interact with the database and retrieve or modify data. Sample :	2	K2	CO1				



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	EXEC SQL BEGIN DECLARE SECTION; int id; char name[20]; EXEC SQL END DECLARE SECTION; EXEC SQL SELECT name INTO :name FROM STUDENT WHERE id = :id;			
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PART B

Q.No.	Questions	Marks	BT L	CO														
1.	Explain the basic architecture of a Database Management System with a neat block diagram	16	K2	CO1														
2.	Discuss in detail about the various Data Models available for data structuring with suitable examples.	16	K2	CO1														
3.	Describe fundamental relational algebra operations and illustrate each with examples	16	K2	CO1														
4.	Demonstrate with necessary syntax and examples the use of DDL, DML, DCL, and TCL commands .	16	K3	CO1														
5.	Consider a student registration database comprising of the below given table schema. Student File <table><tr><td>Student Number</td><td>Student Name</td><td>Address</td><td>Telephone</td></tr></table> Course File <table><tr><td>Course Number</td><td>Description</td><td>Hours</td><td>Professor Number</td></tr></table> Professor File <table><tr><td>Professor Number</td><td>Name</td><td>Office</td></tr></table> Registration file <table><tr><td>Student Number</td><td>Course Number</td><td>Date</td></tr></table> Consider a suitable sample of tuples/records for the above mentioned tables and write DML statements (SQL) and relational algebra equations to solve the queries listed below.	Student Number	Student Name	Address	Telephone	Course Number	Description	Hours	Professor Number	Professor Number	Name	Office	Student Number	Course Number	Date	16	K3	CO1
Student Number	Student Name	Address	Telephone															
Course Number	Description	Hours	Professor Number															
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	1. Which courses does a specific professor teach? and where is his/her office? 2. For a specific student number, in which courses is the student registered and what is his/her name? 3. Who are the professors for a specific student? 4. Who is the student registered in a specific course?			
6.	Assume the following table. • Degree(degcode, name, subject) • Candidate(seatno, degcode, semester, month, year, result) • Marks(seatno, degcode, semester, month, year, papcode, marks) Note: Degcode-degree code, Name-name of the degree (MSc, M.com) Subject-subject of the course. E.g. Phy, Papcode-Paper code E.g. A1 Solve the following queries using SQL: (i) Write a SELECT statement to display all the degree codes which are there in the candidate table but not present in degree table in the order of degcode. (ii) Write a SELECT statement to display the name of all the candidates who have got less than 40 marks in exactly 2 subjects. (iii) Write a SELECT statement to display the name, subject and number of candidates for all degrees in which there are less than 5 candidates. (iv) Write a SELECT statement to display the names of all the candidates who have got highest total marks in MSc.,(Maths)	16	K3	CO1
7.	Demonstrate how Embedded SQL and Dynamic SQL works with necessary examples.	16	K3	CO1
8.	Illustrate the advanced features of SQL with suitable examples.	16	K2	CO1

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